

6.2 The sources, origins, physical and working properties of natural, synthetic, woven and non-woven, knitted, blended and mixed-fibre textiles and their social and ecological footprint	To apply knowledge and understanding of the advantages, disadvantages and applications of the following materials, in order to be able to discriminate between them and select appropriately.
	6.2.1 Natural: - a animal i. wool (in topic 1) ii. silk - b vegetable i. cotton (in topic 1) ii. linen.
	6.2.2 Synthetic: - a polyester (in topic 1) - b acrylic (in topic 1) - c regenerated cellulosic – viscose, acetate, Tencel® - d polyamide - e elastane - f nylon.
	6.2.3 Woven: - a plain – calico (in topic 1) - b twill – denim (in topic 1) - c satin – jacquard - d pile – velvet.
	6.2.4 Non-woven: - a felted wool fabric (in topic 1) - b bonded fibres/webs (in topic 1).
	6.2.5 Knitted: - a weft-knitted fabrics (in topic 1) - b warp-knitted fabrics (in topic 1).

TEXTILES

Key idea	What students need to learn:
	6.2.5 Sources and origins – where natural, synthetic, woven and non-woven, knitted, blended and mixed-fibre textiles are resourced/manufactured and their geographical origin: - a China, India, USA, Pakistan – cotton - b China, India, Uzbekistan – silk - c Russia, Canada, Ukraine, Europe (France and Belgium) – (flax) linen - d Australia, New Zealand, China, USA, United Kingdom – wool - e Alpine Forests – cellulose and wood pulp – cotton linters, pine, spruce, or hemlock trees – soft wood – acetate - f European Forest – oak and birch – hard wood – Lyocell™ - g Russia, UAE, Saudi Arabia – crude oil – polyester, nylon, acrylic.
	6.2.6 The physical characteristics of each natural, synthetic, woven and non-woven, knitted, blended and mixed-fibre textile: - a allergenic - b texture - c density.
	6.2.7 Working properties – the way in which each material behaves or responds to external sources: - a elasticity (in topic 1) - b resilience (in topic 1) - c durability (in topic 1) - d tensile strength - e breathability and absorbency - f electrical conductivity - g heat conductivity.
	6.2.8 Social footprint: - a trend forecasting - b impact of material production on communities and wildlife - c impact of farming and material production on communities and wildlife - d recycling/disposal – ethical responsibility - e reduction of chemical finishes – surface and aftercare treatments - f reduction of packaging materials – reduction in litter/waste/energy use - g brand identity – consumerism, changing the packaging of products over time.

TEXTILES

Key idea	What students need to learn:
	6.2.9 Ecological footprint: - a sustainability - b processing - c transportation - d wastage - e pollution - f deforestation - g oil exploration and extraction - h wildlife loss.